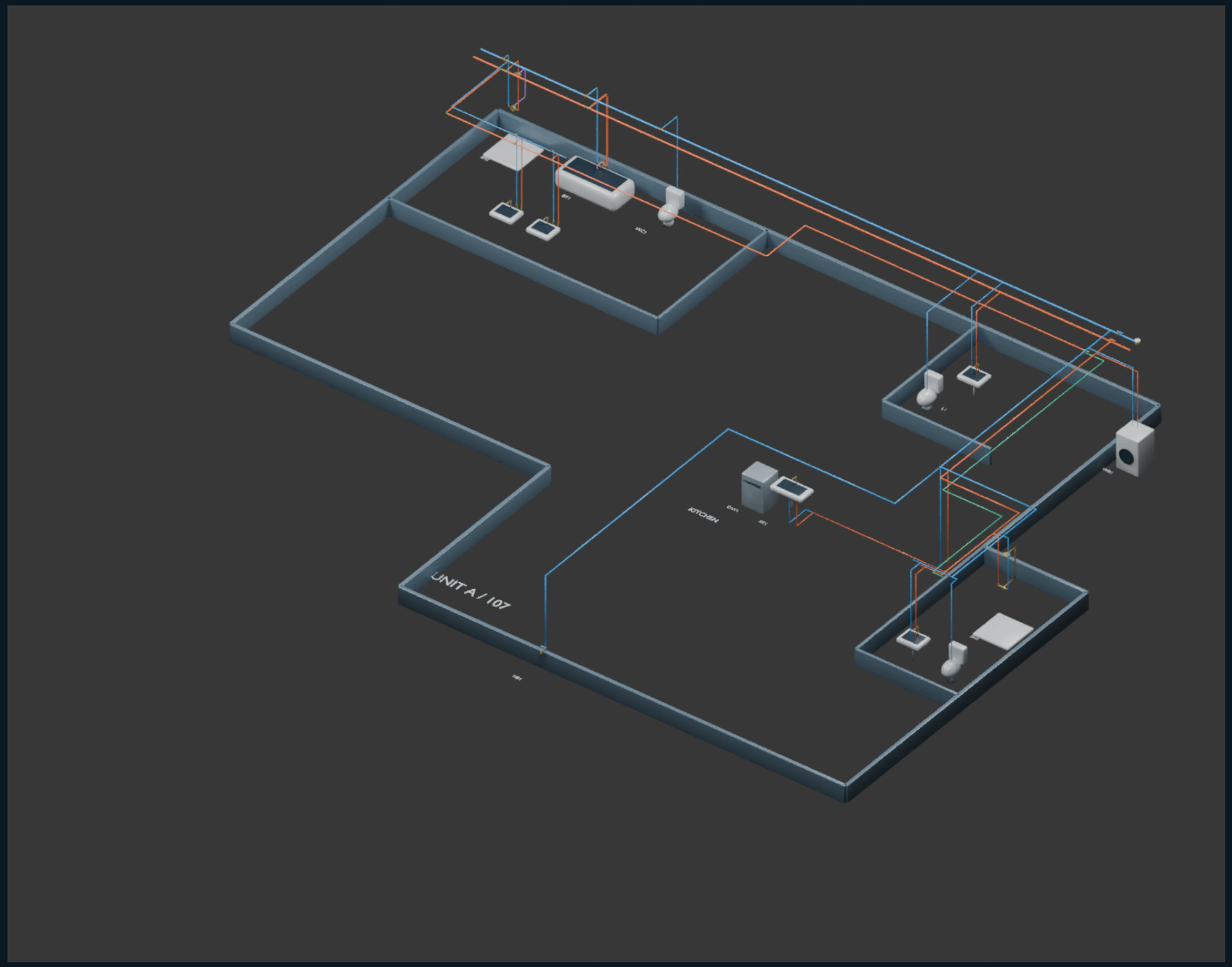
Unit 107: scale tied to the construction plans

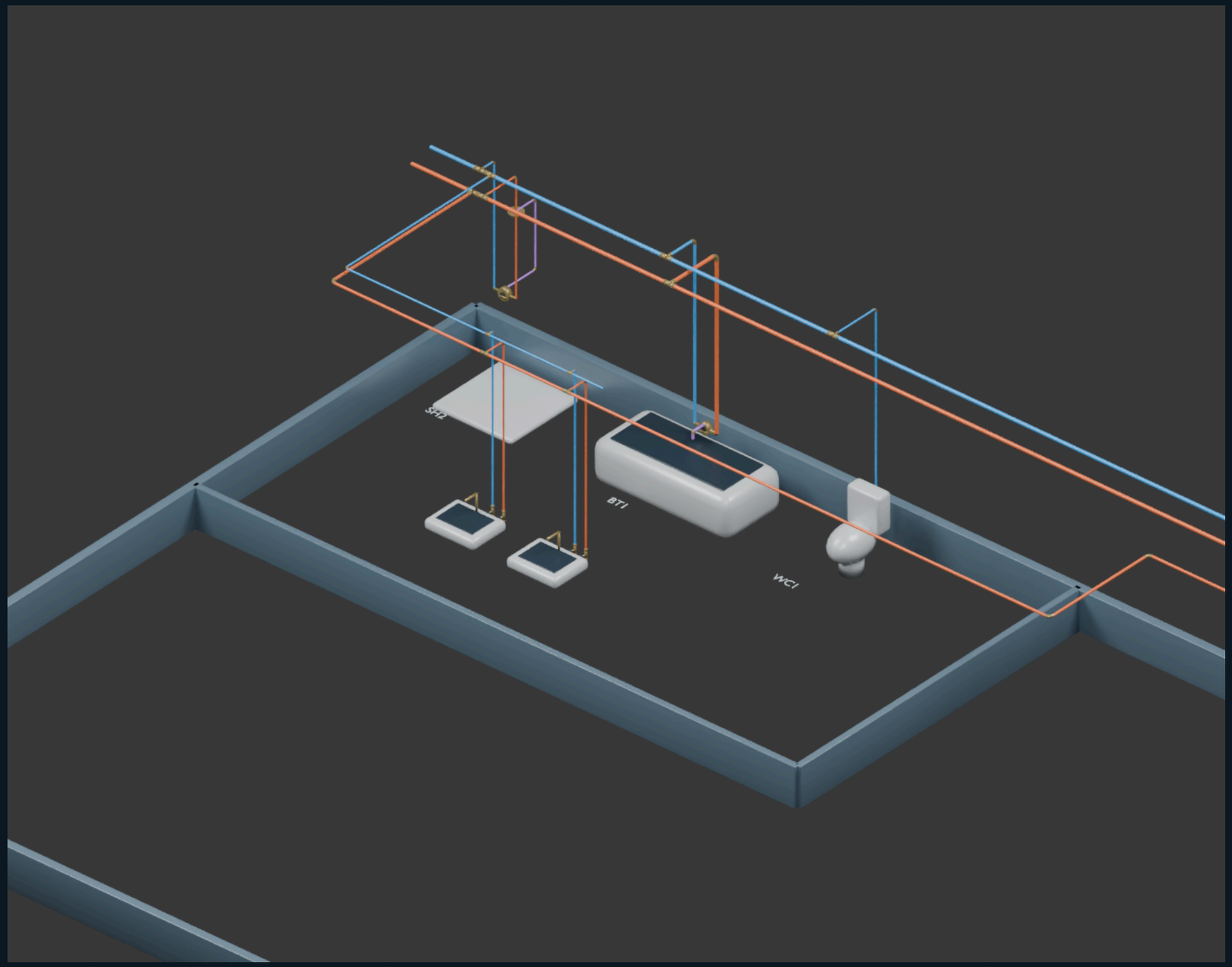
Unit XY is now provisionally calibrated through A2.03 structural grid dimensions and the P2.02 plumbing plan. Colors: blue cold, orange hot, green return, purple mixed water. Pipe heights and fixture envelopes remain assumed.



MODEL STUDY | Unit XY provisionally calibrated; pipe heights and envelopes assumed. Not an as-built.

Primary bathroom: supply connections

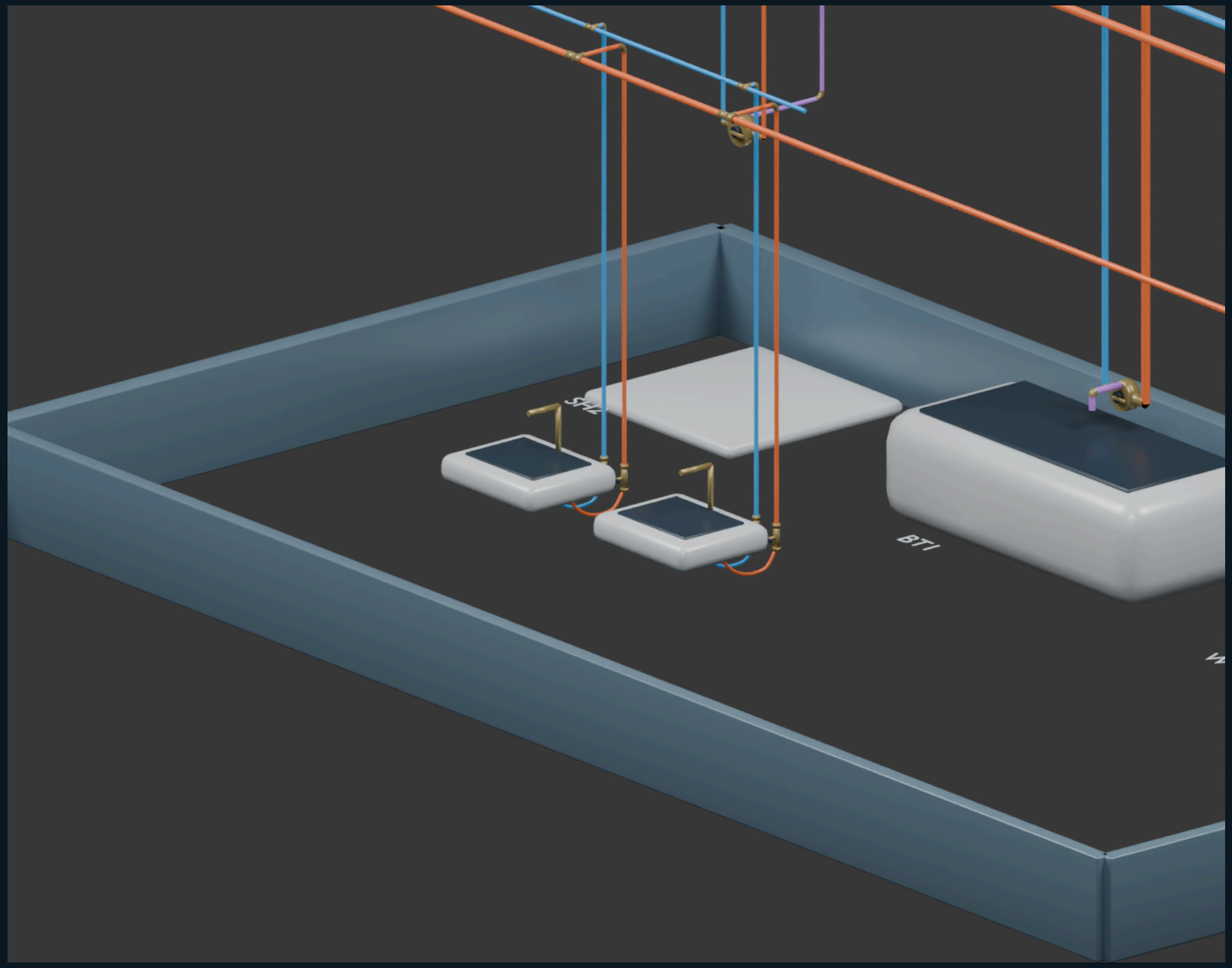
Nine explicit fixture ports connect the two lavatories, tub, shower and toilet. Rounded elbows, branch tees, five shutoffs and two mixing bodies are representative hardware. Heights and local wall-side offsets remain assumed.



MODEL STUDY | Unit XY provisionally calibrated; pipe heights and envelopes assumed. Not an as-built.

Fixture connections: behind the basins

Vertical supplies run behind the fixture envelopes. Short flexible connectors join five shutoffs to the two faucets and toilet tank. Both mixing bodies have connected mixed-water outlets. No sanitary or vent piping is represented.

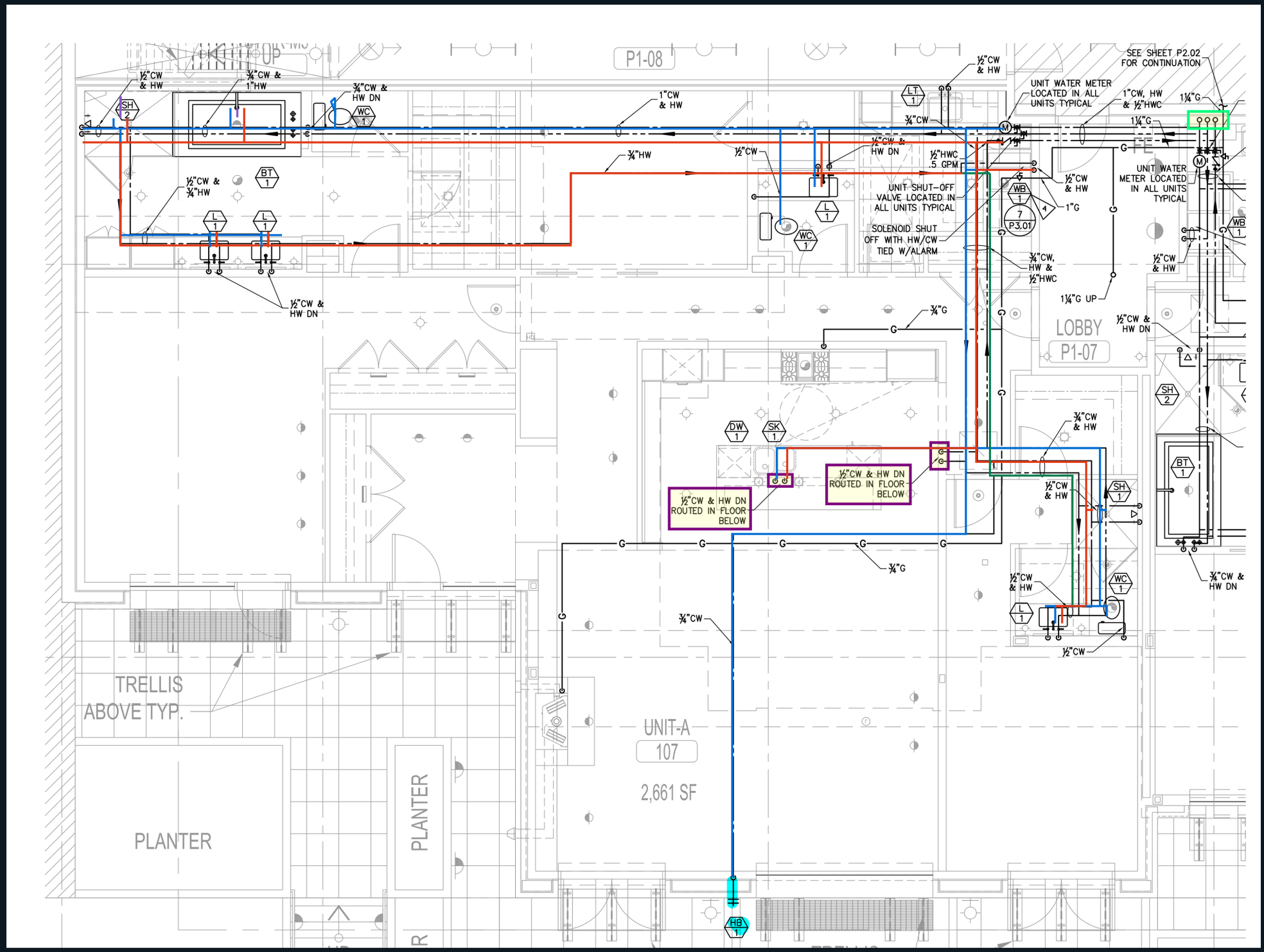


MODEL STUDY | Unit XY provisionally calibrated; pipe heights and envelopes assumed. Not an as-built.

HIGHLAND / DOMESTIC WATER

Check the route before extending the model

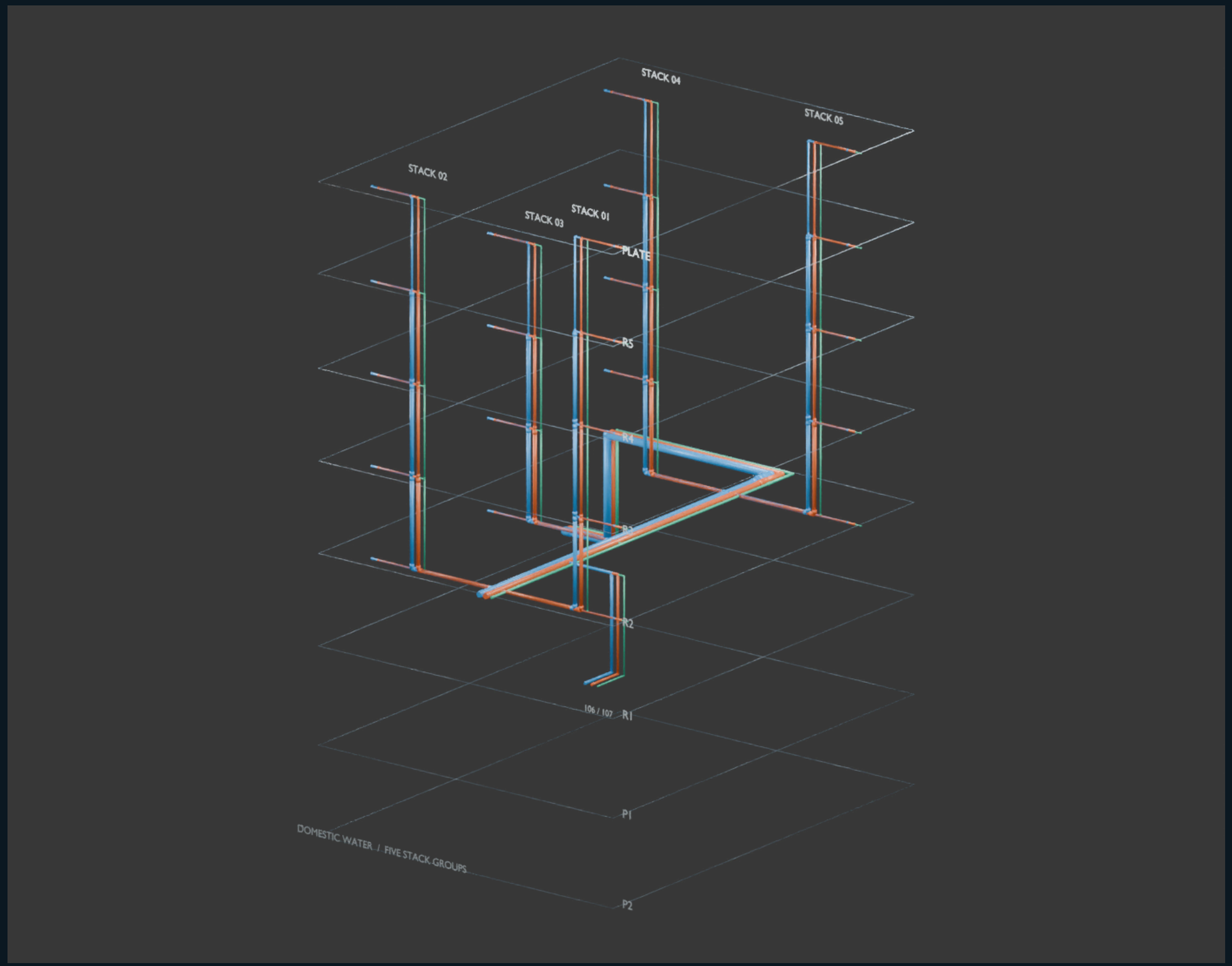
Overlay generated from the calibrated specification and projected back to P4.01. Scale allowance: +/-1%; schematic trace allowance: +/-6 in. Bathroom wall-side service offsets are additional assumptions. Kitchen underfloor path and exact loop joins remain assumptions. Not a field survey.



MODEL STUDY | Unit XY provisionally calibrated; pipe heights and envelopes assumed. Not an as-built.

The shared network: documented floor datums

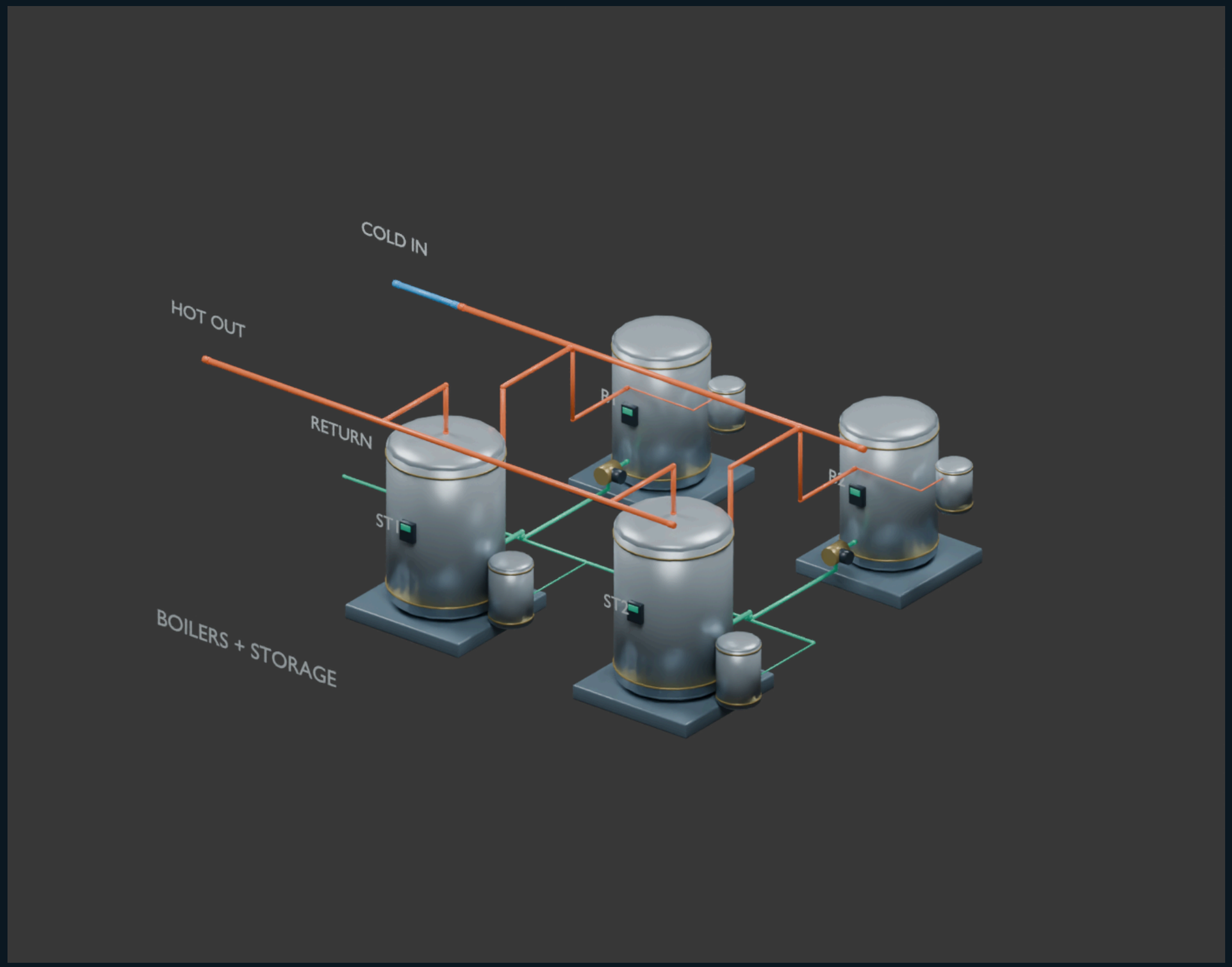
Floor levels now follow A3.11: P2 332.00, P1 342.00, R1 353.00, R2 363.25, R3 373.50, R4 383.75, R5 394.25 ft. The upper guide is PLATE 404.42, not a flat roof. Stack XY and pipe offsets remain assumed.



MODEL STUDY | Unit XY provisionally calibrated; pipe heights and envelopes assumed. Not an as-built.

Plant connections

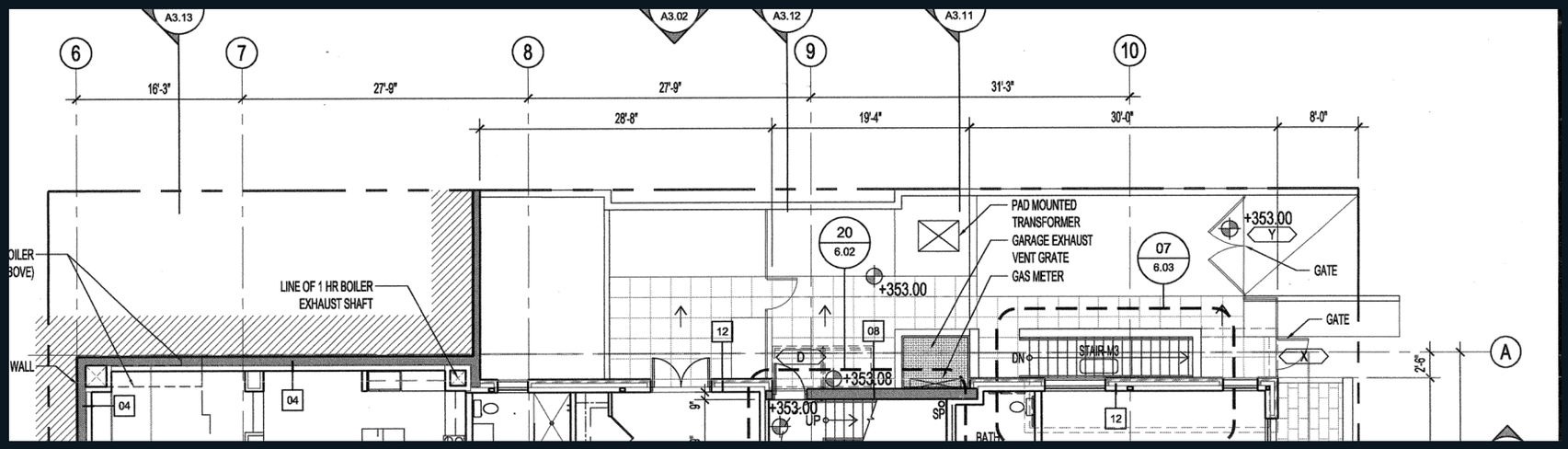
P3.02 / detail 1: two boilers, two storage tanks, four expansion vessels and three installed pumps. WAC2 is a loose spare. Equipment envelopes are representative; relief piping, flues, detailed controls and balancing bypasses are incomplete.



MODEL STUDY | Unit XY provisionally calibrated; pipe heights and envelopes assumed. Not an as-built.

Calibration evidence

A2.03 dimensions the shared structural grid. Those dimensions calibrate P2.02; matched architectural features then transfer scale to the enlarged P4.01 Unit A drawing. The excerpt does not include the P1 architectural plan.

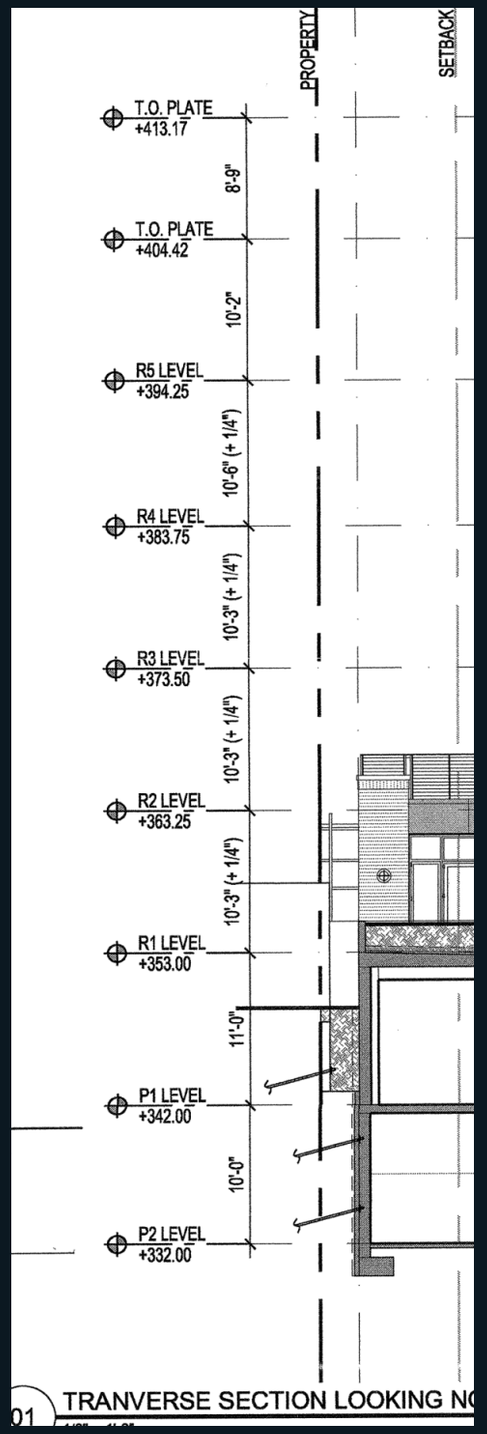


HORIZONTAL GRID
6-7: 16 ft 3 in | 7-8: 27 ft 9 in
8-9: 27 ft 9 in | 9-10: 31 ft 3 in
Overall: 1,236 in across 927 PDF points on P2.02
Scale: 0.75 PDF points / inch

INDEPENDENT VERTICAL CHECK
E-F: 18 ft | F-G: 24 ft 6 in | G-H: 15 ft
All individual spans agree with the working scale within 0.6%.

CROP REGISTRATION
21 matching features; median residual 0.224 PDF points.
Unit conversion: 0.51398 in per normalized pixel.
Plan dimensions increase 28.5% from the earlier placeholder.

DATUM CAUTION
Use the labeled elevations; some interval notes also say (+1/4 in). That small discrepancy is retained for review.
11 ft at Unit 107 is floor-to-floor, not clear ceiling height.



What is checked, and what is still needed

The new construction plans establish building levels and support a provisional horizontal calibration. Concealed pipe placement and equipment geometry still require further evidence.

CHECKED

- Unique IDs and provenance on 360 elements; distribution graphs remain connected.
- Five stack groups, 23 explicit branch groups, 14 Unit A fixtures.
- Bathroom: 9 fixture ports, 5 stops, 2 mixing bodies and 2 mixed outlets checked.
- Seven floor datums match A3.11; Unit 107 floor-to-floor height is 132 in.
- Horizontal and vertical structural-grid spans agree within the raster allowance.
- Cross-sheet registration and model-projected drawing overlay reviewed.

STILL NEEDED

- Unit 107 / P1 architectural and reflected-ceiling plans (not in this excerpt).
- Clear ceiling heights, concealed pipe elevations, offsets and actual shaft positions.
- Equipment dimensions; resolution of the 04 stack R1 and dishwasher connections.
- Clarification of quarter-inch interval notes versus the labeled floor elevations.
- Field records or measurements if installed conditions are the objective.

CURRENT SCOPE

Domestic water supply and return only; sanitary, vent and gas are not modeled.
Working allowances are not survey accuracy: +/-1% scale; +/-6 in local route trace.
No construction lengths or quantities should be taken from this model.